

NITHIN P

✉ nithinpurushothama@gmail.com

☎ +91-9353279658

🌐 linkedin.com/in/nithin-purushothama-70664727b

🐙 github.com/chennakeshavadasa

EDUCATION

- **Visvesvaraya Technological University** Bengaluru, India
Bachelor of Engineering – Electronics and Communication Engineering; CGPA: 8.6/10 Dec 2021 – June 2025

SKILLS SUMMARY

- **Core Skills:** Analog IC Design, Switched-Capacitor Amplifiers, Ring Amplifiers, LDOs, BGRs, OTAs, CMOS Circuit Design, Tapeout (DRC, LVS, PEX), Mixed-Signal Simulation, Digital Signal Processing.
- **EDA Tools:** Cadence Virtuoso/Spectre, NGSpice, Xschem, Magic VLSI, KLayout, Netgen, LTspice, Keysight ADS.
- **PDKs:** TSMC 28 nm, UMC 55 nm, IHP SG13G2 (BiCMOS), SKY130, GF180MCU.
- **Languages:** Python, MATLAB, C/C++, Verilog, SystemVerilog, OCEAN Script, NGSpice Control Language, $\LaTeX$.
- **ML & Quant:** NumPy, Pandas, SciPy, Scikit-learn, PyTorch (basics); Financial modelling, Monte Carlo simulation, Time-series analysis (ARIMA, Holt-Winters), Risk metrics (VaR, CVaR, Sharpe/Sortino).

EXPERIENCE

- **Omni Design Technologies** Bengaluru, India
Analog Design Intern March 2026 – Present
 - **Block Designs:** Designed the **Bandgap Core** and **Reference Analog** blocks for the PVT Monitor and Reference IPs in **TSMC 28 nm CMOS**.
 - ◆ Replaced the BG_CORE opamp with a self-biased folded-cascode OTA and a new always-on startup circuit, fixing a common-mode bug that latched the bandgap loop at VDD; reduced 8-bit I-DAC current mismatch to downstream SC-amp and ADC blocks.
 - ◆ Owned REF_ANALOG: 5 temperature-compensated high-PSRR reference voltages and an $8 \times 200 \mu\text{A}$ current reference, adding trim-based programmability to achieve $\pm 20 \text{ ppm}/^\circ\text{C}$ drift across corners.
 - ◆ Mapped digital infrastructure from a 16 nm reference IP onto the redesigned 28 nm blocks, enabling block-level reuse and accelerating IP closure.
- **PhyTau Semiconductors** Bengaluru, India
Analog Design Engineer Oct 2025 – Feb 2026
 - **Block Designs:** Worked on a **4-input PWM comparator with hysteresis** for a multi-phase DC-DC converter SoC in **UMC 55 nm CMOS**.
 - ◆ Redesigned the comparator to introduce programmable hysteresis at the first stage, eliminating trip-point glitches and reducing propagation delay; reduced power consumption by 30%.
 - ◆ Debugged digital-induced glitches, identifying delay contributions from hysteresis control and de-glitching logic.
- **Indian Institute of Technology Gandhinagar (IITGN)** Gandhinagar, India
Research Intern (Advisor: Prof. Madhav K. Pathak, LP-CAS Lab) Feb 2025 – Aug 2025
 - **Block Designs:** Dynamic Amplifiers and Ring Amplifier-based LDO at LP-CAS Lab, IITGN.
 - ◆ Developed a **design methodology** for non-linear dynamic amplifiers modelling RC, large-signal, and small-signal settling stages; validated on an inverter-based **switched-capacitor amplifier**.
 - ◆ Proposed an LDO architecture where the **Ring amplifier final stage serves as the pass transistor**, using adaptive biasing and a deadzone regulation circuit for robust PVT performance.
 - ◆ Proposed a **translinear circuit-based MAC architecture** for analog neural networks using a digitally controlled subthreshold MOSFET cell with a current-steering DAC for programmable neuron weights.
 - ◆ Served as **Teaching Assistant (TA)** for the graduate course *PMIC Design (EE660)*.
 - **Tapeout 🏠:** Designed a **piecewise-compensated, high-PSRR Bandgap Reference** with sub-5 ppm/ $^\circ\text{C}$ TC in SKY130, including a low-power always-on startup circuit; fabricated through the **TinyTapeout SKY130** shuttle.
- **Indian Space Research Organisation (ISRO)** Hassan, India
Intern Oct 2023 – Nov 2023
 - **Project:** GeoSat-based satellite communication in the Antenna System for Satellite TTC within C-Band (6 GHz) at the Baseband RF Division, Master Control Facility. Calibrated antenna systems to achieve Cross-Polar Discrimination (XPD) during In-Orbit Testing of GSAT-31.

PUBLICATIONS & PRESENTATIONS

- **R_{on}/g_m Based Design Methodology for Dynamic Amplifiers** ESSERC 2026 (Upcoming)
Nithin P, Pramoda SR, Suyajnaa, Runpeng Gao, Praveen Kumar Venkatachala, Prof. Madhav Pathak
IEEE SSCS Code-a-Chip Travel Grant Award winner. Poster presentation at the 52nd European Solid-State Electronics Research Conference (ESSERC 2026), Palma de Mallorca, Spain, September 2026.
- **Analog CMOS Multiplier Circuits: Design and Comparative Analysis** IEEE ICCE 2025
Nithin P, Prof. Madhav Pathak
Design and comparison of analog multiplier circuits for low-power pre-ADC sensor processing, power management, and analog neural networks using translinear and linear MOSFET-based MAC architectures.

- IEEE SSCS “Code-a-Chip” Travel Grant Award**  **2026**
 IEEE Solid-State Circuits Society (SSCS) – Open-Source Ecosystem (TC-OSE)
Awarded \$5,000 travel grant (Young Professional’s category) for “ R_{on}/g_m Based Design Methodology for Dynamic Amplifiers” a notebook demonstrating a novel design methodology for inverter-based dynamic amplifiers on IHP SG13G2 using open-source EDA tools. Award ceremony at ESSERC 2026, Palma de Mallorca, Spain.

PROJECTS

- Career Semi-Quant Terminal** : Zero-backend web application providing institutional-grade quantitative analysis of 40+ semiconductor companies. Pure-JavaScript analytics engine implementing RSI, MACD, Bollinger Bands, VaR/CVaR, Beta, Sharpe/Sortino ratios, and GBM-based Monte Carlo simulations with automated multi-source financial data aggregation.
- Design of Power-on Reset Circuit** : Designed a POR circuit in SKY130 PDK using a current comparator with PMOS/NMOS current mirrors for threshold detection; V_{POR} asserts Enable at $V_{DD} = 1.5\text{ V}$ across PVT corners.
- Automation of PVT and SPICE Simulation Framework for Analog Design** : Open-source Python tools for analog IC design automation: PVT sweep framework for NGSpice with automated metric extraction and comparison plots; SPICE-to-area estimation tool using Magic VLSI for early floorplanning before physical layout.
- Design of Process-Invariant Ring Amplifier using Deadzone Regulation** : Designed a PVT-robust Ring amplifier in IHP SG13G2 (BiCMOS). Deadzone regulation circuit with negative feedback controls third-stage quiescent current across corners. Dynamic error analysis covers RC, large-signal, and small-signal settling regimes; validated in a switched-capacitor amplifier.
- Ultra-Low Power Two-Transistor Subthreshold Voltage Reference** : Voltage reference using native NMOS devices matching subthreshold currents between a diode-tied NMOS and a load transistor. Achieved 60 ppm/°C TC across process corners with power $\leq 2\text{ pW}$.
- Piecewise-Compensated High-PSRR Bandgap Reference** : Bandgap reference with higher-order curvature correction in SKY130 and GPDK90; 4.4 ppm/°C TC using piecewise BJT non-ideality compensation and improved PSRR via cascode current mirrors. Full PVT corner verification.
- Low Dropout Voltage Regulator (LDO)** : LDO in SKY130 achieving 1 μs transient settling for a 100 pF, 50 mA load with -60 dB PSRR. Integrated a digital on-chip load for transient testing and an enable switch for complete shutdown.
- Rail-to-Rail Input Swing OTA** : Folded-cascode OTA in SKY130 with complementary PMOS/NMOS differential pair folding for rail-to-rail swing; 63–64 dB DC gain, 2 MHz UGF, 85° phase margin across all PVT corners; input-referred offset $\mu = 0.11\text{ mV}$, $\sigma = 1.54\text{ mV}$.
- 2-Stage Fully Differential OTA** : Fully differential OTA in SKY130 with PMOS differential pair and NMOS load. Buffer-based common-mode reference stabilises output CM level for high gain and bandwidth.
- Folded-Cascode OTA Tapeout** : Designed and submitted a Folded-Cascode OTA in GF180MCU-D PDK through the TinyTapeout GF180 (TTGF0P3) analog shuttle. Completed full pre-tapeout verification including DRC, LVS, and PEX extraction.